EX.1 CRUD WITH UI

```
import tkinter as tk
from tkinter import ttk, messagebox
import mysql.connector


# ---------------- MYSQL CONNECTION ----------------

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="",
    database="studentdb"
)

cursor = conn.cursor()


# ---------------- TKINTER WINDOW ----------------

root = tk.Tk()

root.title("Student CRUD Application")

root.geometry("900x550")


selected_id = None


# ---------------- FUNCTIONS ----------------

def clear_fields():

    firstname_entry.delete(0, tk.END)
    lastname_entry.delete(0, tk.END)
    mobile_entry.delete(0, tk.END)
    email_entry.delete(0, tk.END)


def display_data():

    for item in tree.get_children():
        tree.delete(item)

    cursor.execute("SELECT * FROM students")

    rows = cursor.fetchall()

    for row in rows:
        tree.insert(
            "",
            tk.END,
            values=row
        )


def insert_data():

    firstname = firstname_entry.get()
```

```
    lastname = lastname_entry.get()
    mobile = mobile_entry.get()
    email = email_entry.get()

    if firstname == "" or lastname == "":

        messagebox.showerror(
            "Error",
            "Please enter First Name and Last Name"
        )

        return

    query = """
        INSERT INTO students
        (firstname, lastname, mobile, email)

        VALUES (%s, %s, %s, %s)
    """

    values = (
        firstname,
        lastname,
        mobile,
        email
    )

    cursor.execute(
        query,
        values
    )

    conn.commit()

    messagebox.showinfo(
        "Success",
        "Student Added Successfully"
    )

    clear_fields()

    display_data()


def select_data(event):

    global selected_id

    selected = tree.focus()

    if selected:

        values = tree.item(
            selected,
            "values"
        )

        selected_id = values[0]

        clear_fields()
```

```
        firstname_entry.insert(
            0,
            values[1]
        )

        lastname_entry.insert(
            0,
            values[2]
        )

        mobile_entry.insert(
            0,
            values[3]
        )

        email_entry.insert(
            0,
            values[4]
        )


def update_data():

    global selected_id

    if selected_id is None:

        messagebox.showerror(
            "Error",
            "Please select a student"
        )

        return

    query = """
        UPDATE students

        SET firstname=%s,
            lastname=%s,
            mobile=%s,
            email=%s

        WHERE id=%s
    """

    values = (
        firstname_entry.get(),
        lastname_entry.get(),
        mobile_entry.get(),
        email_entry.get(),
        selected_id
    )

    cursor.execute(
        query,
        values
    )

    conn.commit()
```

```
    messagebox.showinfo(
        "Success",
        "Student Updated Successfully"
    )

    clear_fields()

    display_data()


def delete_data():

    global selected_id

    if selected_id is None:

        messagebox.showerror(
            "Error",
            "Please select a student"
        )

        return

    confirm = messagebox.askyesno(
        "Confirm Delete",
        "Do you want to delete this student?"
    )

    if confirm:

        query = """
            DELETE FROM students
            WHERE id=%s
        """

        cursor.execute(
            query,
            (selected_id,)
        )

        conn.commit()

        messagebox.showinfo(
            "Success",
            "Student Deleted Successfully"
        )

        clear_fields()

        display_data()


# --------------- FORM ---------------

tk.Label(
    root,
    text="First Name"
).place(
    x=50,
```

```
    y=50
)

firstname_entry = tk.Entry(
    root,
    width=30
)

firstname_entry.place(
    x=180,
    y=50
)


tk.Label(
    root,
    text="Last Name"
).place(
    x=50,
    y=100
)

lastname_entry = tk.Entry(
    root,
    width=30
)

lastname_entry.place(
    x=180,
    y=100
)


tk.Label(
    root,
    text="Mobile"
).place(
    x=50,
    y=150
)

mobile_entry = tk.Entry(
    root,
    width=30
)

mobile_entry.place(
    x=180,
    y=150
)


tk.Label(
    root,
    text="Email"
).place(
    x=50,
    y=200
)

email_entry = tk.Entry(
```

```
    root,
    width=30
)

email_entry.place(
    x=180,
    y=200
)


# --------------- BUTTONS ---------------

tk.Button(
    root,
    text="Add",
    command=insert_data
).place(
    x=180,
    y=250
)


tk.Button(
    root,
    text="Update",
    command=update_data
).place(
    x=250,
    y=250
)


tk.Button(
    root,
    text="Delete",
    command=delete_data
).place(
    x=330,
    y=250
)


tk.Button(
    root,
    text="Clear",
    command=clear_fields
).place(
    x=410,
    y=250
)


# --------------- TREEVIEW ---------------

columns = (
    "ID",
    "First Name",
    "Last Name",
    "Mobile",
    "Email"
)
```

```
tree = ttk.Treeview(
    root,
    columns=columns,
    show="headings"
)


for column in columns:

    tree.heading(
        column,
        text=column
    )


tree.column(
    "ID",
    width=50
)


tree.place(
    x=400,
    y=50,
    width=480,
    height=400
)


tree.bind(
    "<ButtonRelease-1>",
    select_data
)


# ---------------- LOAD DATA ----------------

display_data()


# ---------------- RUN ----------------

root.mainloop()
```